5. Construct a scheduling program with C that selects the waiting process with the highest priority to execute next.

main.c

Run ▶

```c
#include <stdio.h>
int main() {
    int n, i, j, temp;
    int pid[20], priority[20];
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("Enter priority of Process %d: ", pid[i]);
        scanf("%d", &priority[i]);
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;
                temp = pid[i];
                pid[i] = pid[j];
                pid[j] = temp;
            }
        }
    }
    printf("\nExecution Order:\n");
    for (i = 0; i < n; i++) {
        printf("Process %d (Priority %d)\n",
```

Output

```
Enter number of processes: 4
Enter priority of Process 1: 3
Enter priority of Process 2: 1
Enter priority of Process 3: 4
Enter priority of Process 4: 2

Execution Order:
Process 2 (Priority 1)
Process 4 (Priority 2)
Process 1 (Priority 3)
Process 3 (Priority 4)
```